
Toscanini 01 executive technology brief

inv:fixture:fluidics_hydraulic0e8d4aa7cfb8 · run r541-ranked-pkg-battery-a · maturity
EARLY_TECHNICAL_EVALUATION · 3D class SYSTEM_3D

Technology

inv:fixture:fluidics_hydraulic:0:e8d4aa7cfb8

Bottleneck

- Problem Id: fixture:fluidics_hydraulic:0
- Device: CSF shunt system
- Failure: proximal catheter obstruction by tissue
- Failure Mode: OBSTRUCTION
- Constraint: maintain long-term lumen patency and flow

Causal mechanism

- Mechanism: porous microstructure resists fluid-path tissue ingrowth. The proposed intervention realizes this mechanism by placing porous titanium proximal catheter tip in the shunt lumen at the failure site identified in the problem statement, where the governing physical effect produces sustained...

Differentiating architecture

- Collision Novelty Risk: SEARCHED_NO_DIRECT_TITLE_MATCH
- Prior Art Status: NO_MATCH_FOUND
- Multi Source Directions: (empty record)

Expected technical advantage

UNKNOWN (not recorded)

Strongest supporting evidence

Evidence classification counts from the run's own verification: direct support: 0; partial support: 0; background: 0; analogy: 1; contradictory: 0; irrelevant: 0; direct-support spans are itemized in 04_EVIDENCE_SUMMARY.

Biggest uncertainty

FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-003

Engineering status

EARLY_TECHNICAL_EVALUATION; 9 design inputs, 6 failure modes, 3 governing equations recorded

Decisive experiment

physical bench/clinical experiment — NOT simulation

Buyer decision

evaluate this package against the nine questions in 03_BUYER_DECISION_CARD; the cheapest falsification is named there with its acceptance rule

Epistemic boundary: DESIGNED / SIMULATED / EVIDENCE-SUPPORTED / EXPERIMENTALLY VERIFIED states come only from the canonical invention object. EXPERIMENTALLY VERIFIED requires real reality-loop evidence. Potential IP territory — formal legal review required. Toscanini performs technology discovery; patentability, claim drafting, infringement/FTO and legal opinions are matters for qualified IP counsel.